Tracking objects on video using world models

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Problem

OSTrack searches only in Region of interest, and after some occlusions the target reappears outside of the “search box” (RoI), which results in lost target.

Idea

Use V-JEPA 2 to find the target across the whole frame and warn when an occlusion starts, while OSTrack stays the one that verifies and tracks.

Why OSTrack?

Why OSTrack: a strong tracker with the best trade-off of quality and speed, and easy to run on our hardware.



AO (average overlap) is GOT-10k's main metric.

figure from Joint Feature Learning and Relation Modeling for Tracking: A One-Stream Framework

Data and Metrics

Data

LaSOT, 1,400 videos in 70 classes, long videos with occlusions. Our splits:

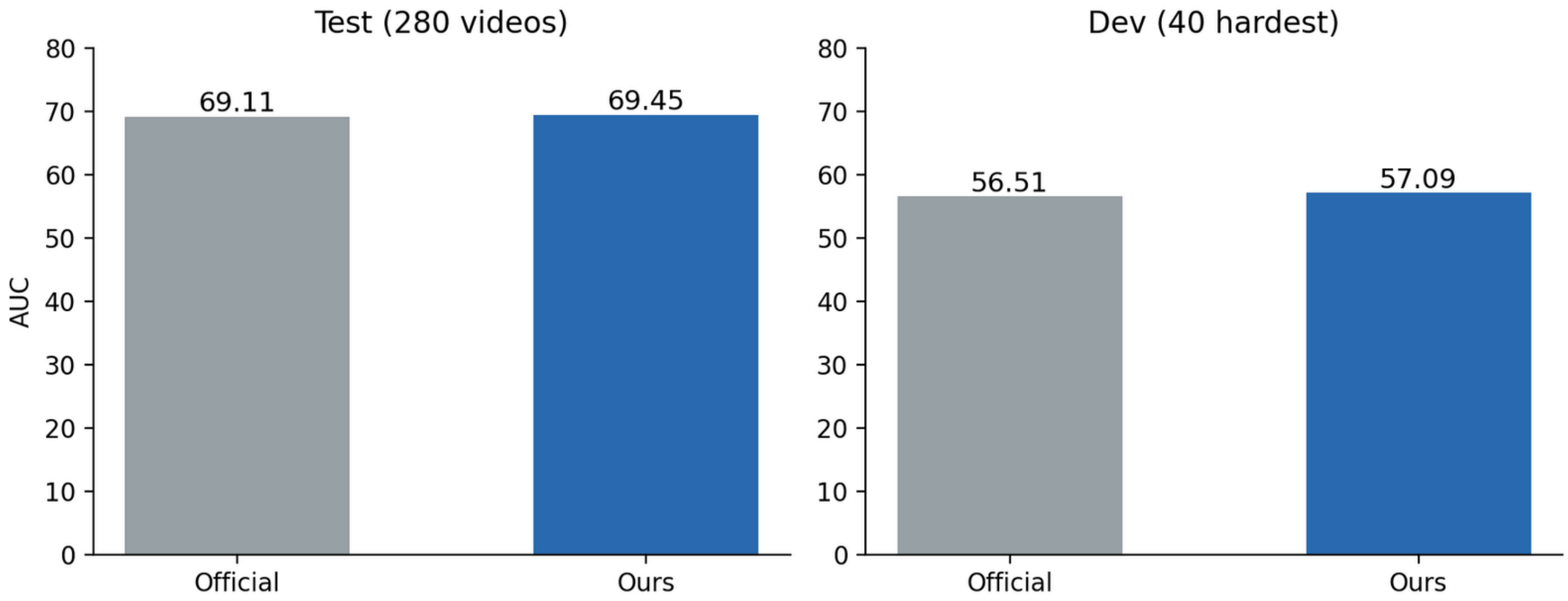
- train: 280 videos
- dev: the 40 hardest test videos (all comparisons run here)
- holdout: the other 240 test videos (final check)

Metrics

- AUC (Area Under Curve): overlap quality of the box.
- P (precision) / NP (normalized precision): center-distance precision, and the scale-normalized version.
- ORR_all (Occlusion Recovery Rate): the share of all occlusions after which the tracker is back on the target (IoU > 0.5 within 30 frames).

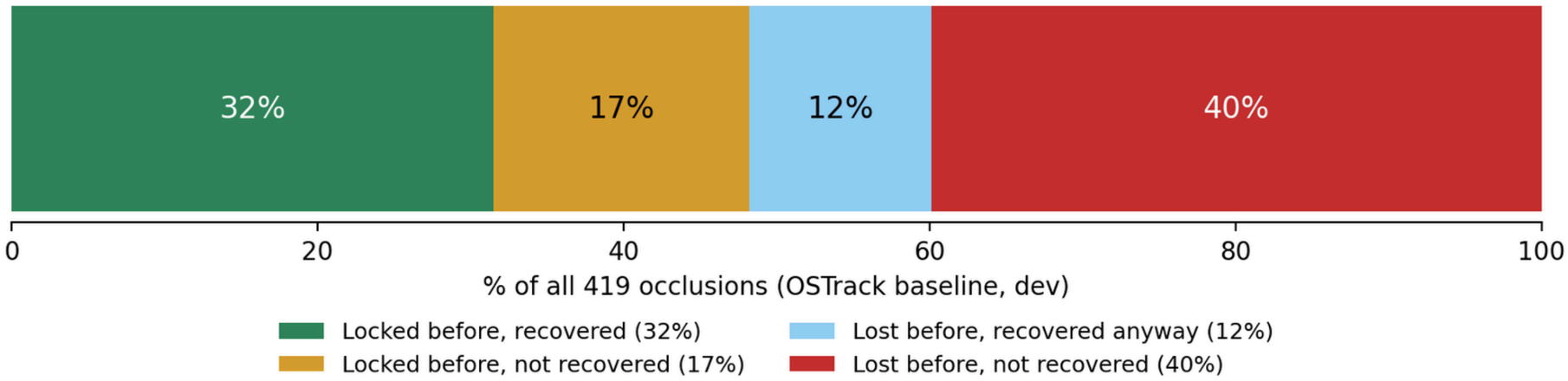
Baseline

Reproduced OSTrack on the **LaSOT** test set



280 videos - LaSOT test dataset
40 hard videos - test videos selected for dense and long occlusions, with at most one video per class

Occlusion Recovery in Baseline OSTrack



Results without predictor

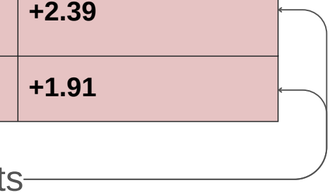
benchmarks

Variant	AUC	Δ AUC	P	NP	ORR	ORR_all	Δ ORR_all
OSTrack (official results)	56.51	−0.58	63.78	64.69	66.50	42.24	−1.19
OSTrack (reproduced)	57.09	0.00	63.51	64.98	65.35	43.44	0.00
Relocation, no verification	55.58	−1.52	61.54	63.14	61.29	40.33	−3.10
Relocation + OSTrack verification	56.39	−0.70	62.69	64.18	63.90	+1.67	+1.67
Relocation + verification, score ≥ 0.6	57.61	+0.51	64.20	65.64	65.87	45.35	+1.91
+ YOLO-World candidates	57.24	+0.15	63.63	65.16	65.02	44.39	+0.95
fusion_v1 (V-JEPA features in the OSTrack crop)	56.92	−0.17	63.36	65.00	65.50	41.53	−1.91
fusion_v1h (+ retrained score head)	57.11	+0.02	63.41	65.19	64.36	42.00	−1.43
fusion_v1h + relocation	57.37	+0.28	63.60	65.35	62.68	43.20	−0.24
fusion_v2a (V-JEPA memory prior in the head)	56.87	−0.23	63.34	64.72	66.19	44.15	+0.72
fusion_v2a + relocation	57.80	+0.70	64.43	65.83	62.91	44.39	+0.95
Score-only warning (control for v1)	56.33	−0.76	62.45	64.10	63.18	44.63	+1.19
Memory guard + drift check	57.48	+0.38	63.97	65.43	64.90	44.39	+0.95
Final, score-only warning	57.00	−0.09	63.40	64.87	64.88	44.87	+1.43
Final without warning (guard + drift + motion)	**57.47**	**+0.38**	**63.93**	**65.37**	**66.99**	**45.58**	**+2.15**

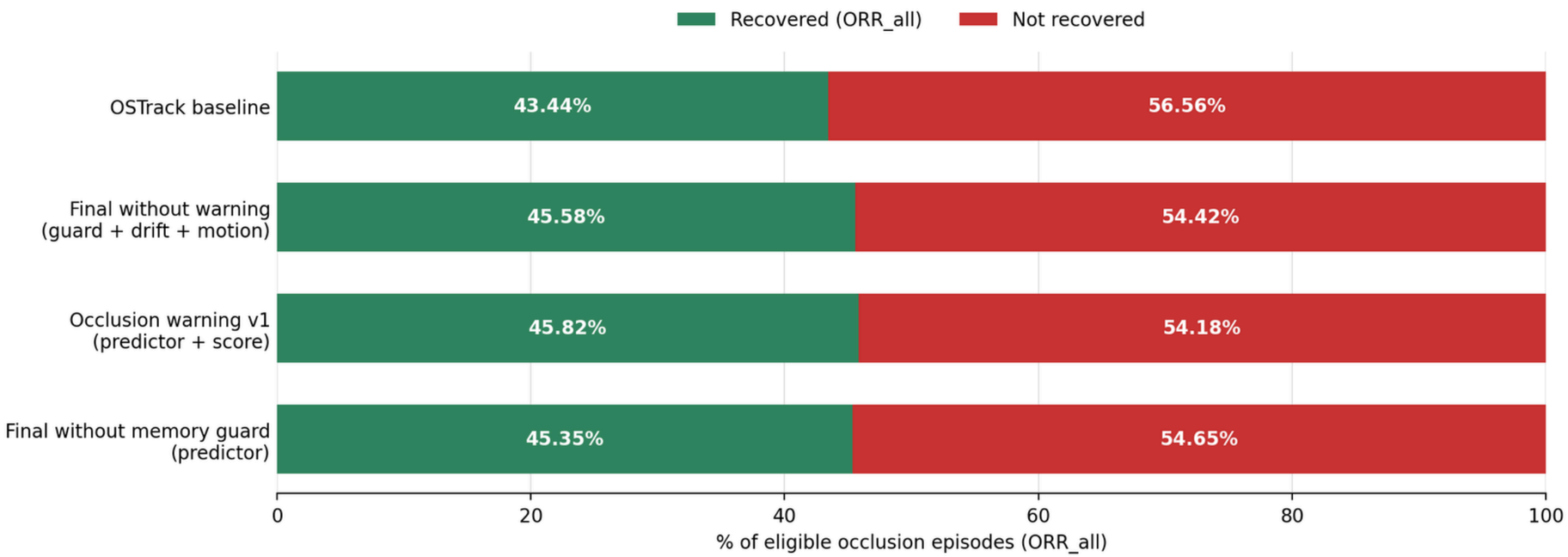
best result

Results with predictor

Variant	AUC	Δ AUC	P	NP	ORR	ORR_all	Δ ORR_all
Surprise as a relocation gate	56.07	−1.03	62.30	63.77	64.47	42.24	−1.19
fusion_v1h + relocation + surprise gate	57.44	+0.35	63.80	65.58	66.67	43.68	+0.24
Final (all parts)	57.49	+0.40	63.93	65.43	65.35	43.91	+0.48
Final without future candidates	57.64	+0.55	64.18	65.62	64.90	45.11	+1.67
Final without drift check	56.92	−0.17	63.29	64.80	61.73	41.77	−1.67
Final without motion prior	57.84	+0.75	64.42	65.89	64.59	45.11	+1.67
Occlusion warning v1 (predictor + score)	**57.00**	**−0.09**	**63.41**	**64.88**	**64.22**	**45.82**	**+2.39**
Final without memory guard	**57.71**	**+0.61**	**64.23**	**65.73**	**67.15**	**45.35**	**+1.91**

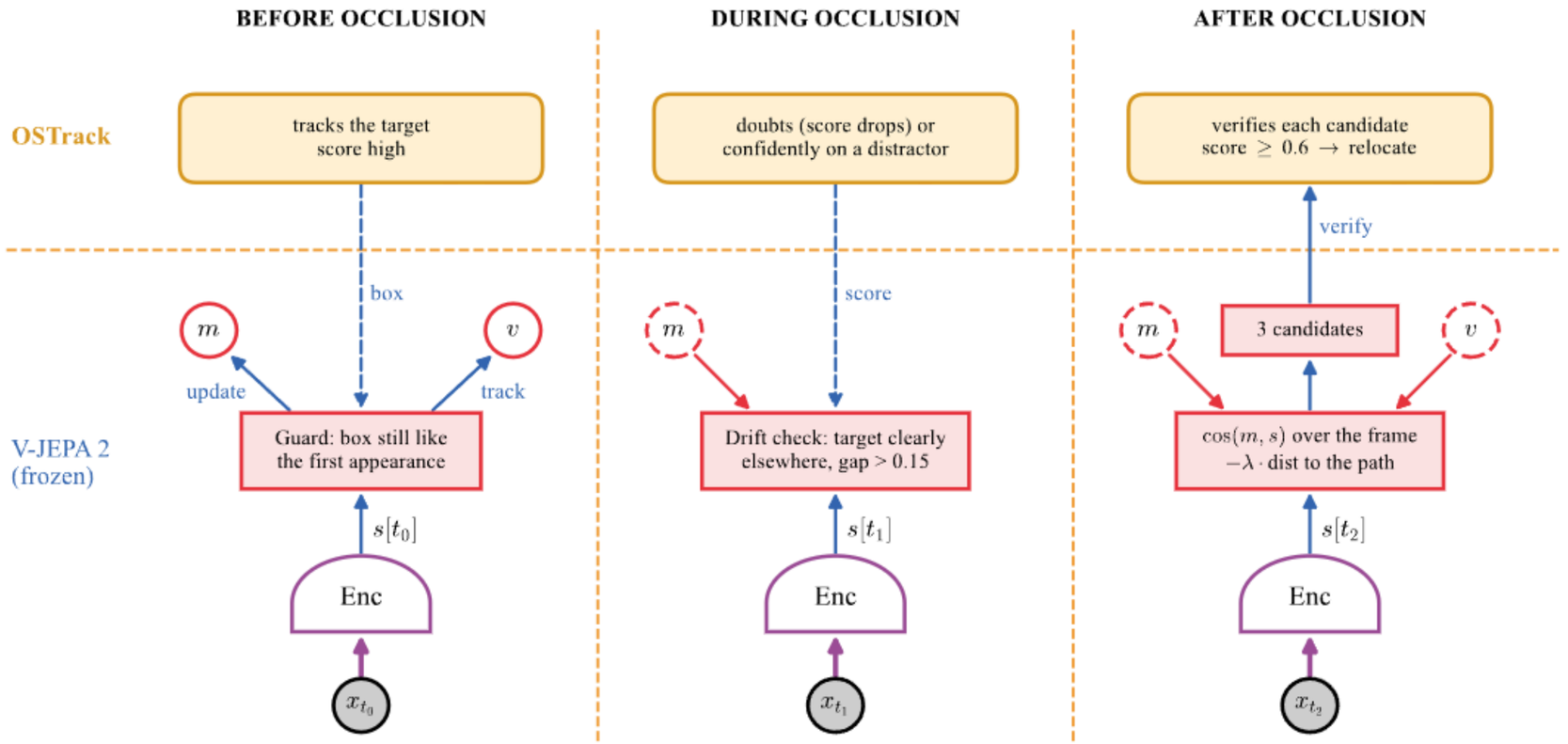


Occlusion Recovery across best models



Best without the predictor

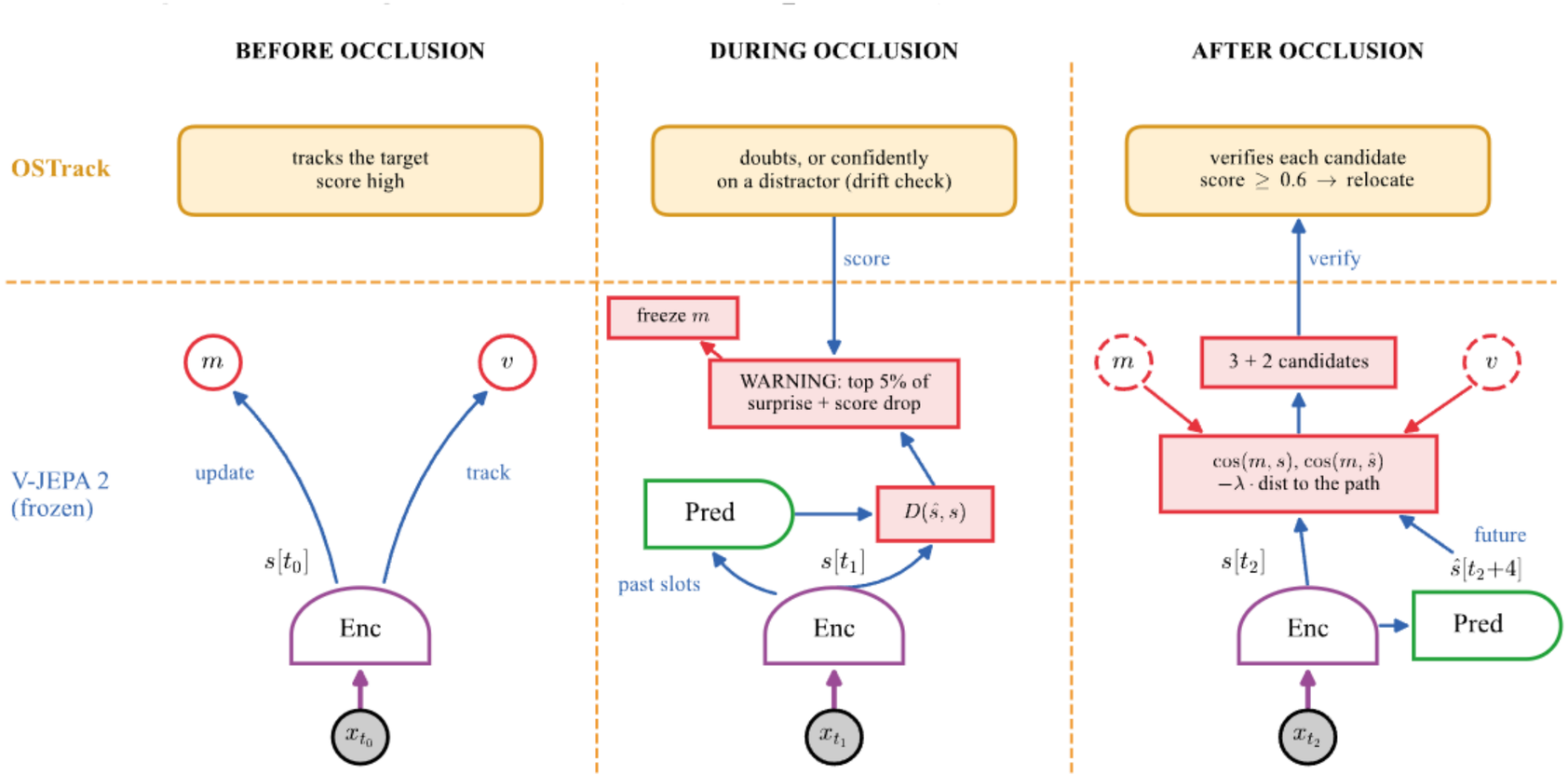
Relocation verified by OSTrack + memory guard + drift check + motion prior (AUC +0.38 ORR_all +2.15)



Dashed circles: state carried from earlier frames (m target memory, v target track). Encoder only, nothing trained.

Best with the predictor: final sys. w/o memory guard

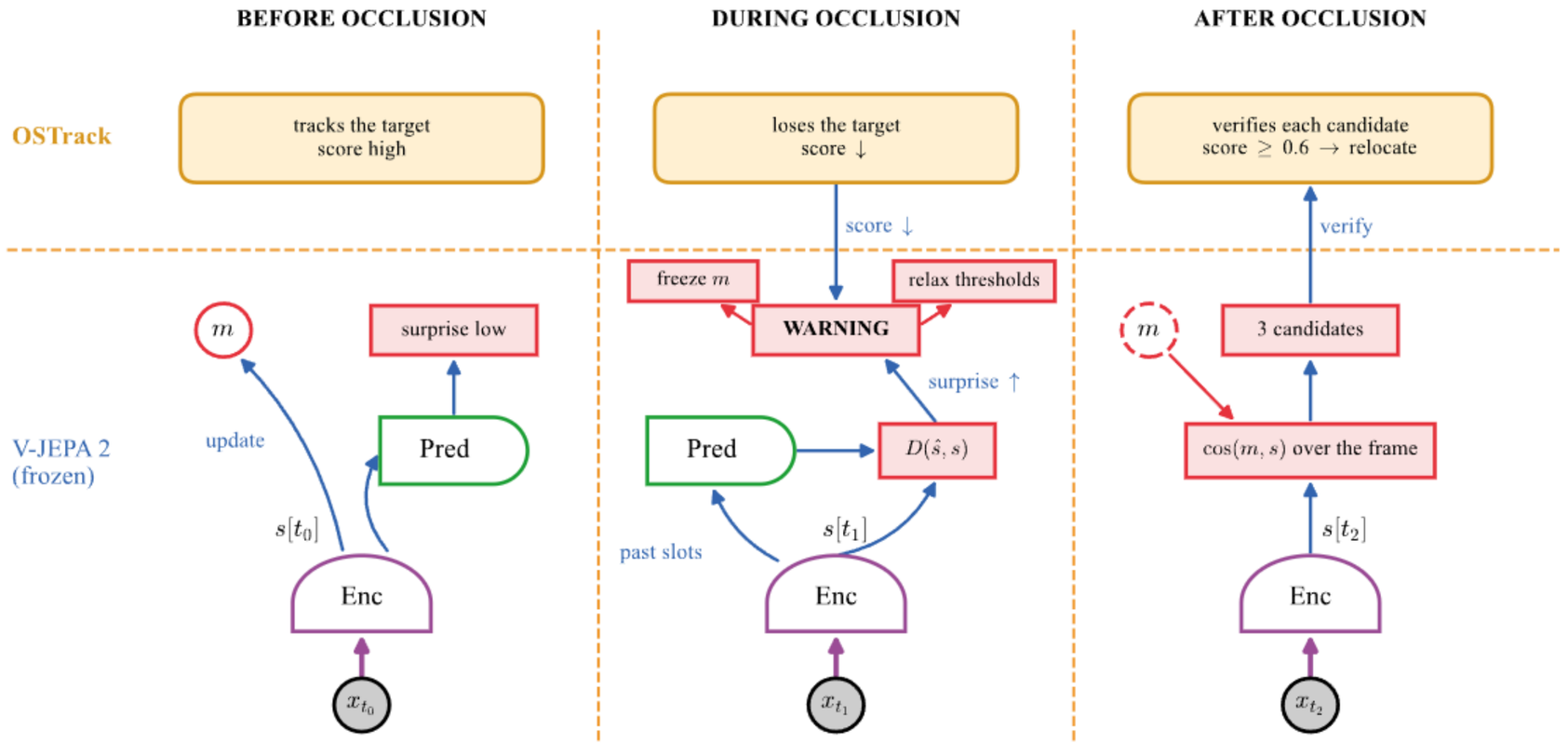
Predictor warning + drift check + motion prior + future candidates
(AUC 57.71, ORR_all 45.35 on dev)



Encoder: target memory m and full-frame search. Predictor: occlusion warning (during) and the predicted future map ŝ (after).

Best ORR_all, with the predictor: occlusion warning v1

Predictor surprise + score drop freeze the memory and relax relocation (AUC 57.00, ORR_all 45.82 on dev)



Pred predicts the current moment from the past frames; D = 1 - cos(predicted, real) at the target. Dashed circle: frozen memory.

Next steps

- Test the best no-predictor and predictor variants on the 240-video holdout split.
- Add bootstrap confidence intervals to measure uncertainty in the recovery results.
- Compare against a wider OSTrack search window and a DINOv2-based candidate search.
- Improve recovery when OSTrack confidently drifts onto a distractor.
- Evaluate the approach on additional tracking datasets.

Summary

V-JEPA as 'eyes on the whole frame' helps find a target that has left the frame: ORR_all +1.9 over the baseline. The V-JEPA predictor is useful as an early warning that an occlusion is starting, and it makes the warning safer than one based on OSTrack's score alone: +0.64 AUC. The main open problem is the tracker confidently drifting onto a distractor before the occlusion even starts: OSTrack, acting as the judge, is fooled itself, so the next step is a judge that takes the world model into account.

References

0.83	0.27	0.43	0.85
0.47	0.55	0.37	0.83
0.85	0.18	0.46	0.50
0.85	0.21	0.24	0.46
0.72	0.35	0.00	0.63

Thank you!